

4. Engineering & UX Decisions

Trade-off Decision

Feature Not Included: Customer Analytics

I decided not to include customer analytics (e.g., customer demographics, repeat customer rate, customer lifetime value) in the dashboard.

Reason

The primary goal of the Command Center is to help merchants quickly understand their daily sales performance and inventory health across Shopify and TikTok Shop. Including customer analytics would increase screen complexity and distract users from the most critical operational metrics.

By focusing on sales, orders, revenue, inventory status, and platform health, the dashboard remains simple, actionable, and aligned with the stated business objective.

Edge Case Handling: TikTok Data Delay or Unavailability

Problem

Shopify provides real-time inventory updates, while TikTok Shop data may be delayed or temporarily unavailable. Displaying missing data without explanation could confuse merchants and reduce trust in the dashboard.

UI Solution

The dashboard communicates this condition through several non-blocking visual indicators:

1. A warning banner at the top of the dashboard:
 - "TikTok inventory data may be delayed. Shopify data remains real-time."
2. Platform status cards:
 - Shopify displays a green "Real-Time" badge.
 - TikTok displays a yellow "Delayed" badge with the last successful sync time.
3. Inventory table indicators:
 - Missing TikTok inventory values are displayed as "N/A".
 - A "Delayed" status badge is shown for affected products.
4. The rest of the dashboard remains fully functional using available Shopify data.

Why This Approach

The design avoids alarming the merchant by using informational warnings rather than error messages. Instead of showing a red failure state, the dashboard clearly explains that TikTok data is temporarily delayed while continuing to display the latest available information.

This approach maintains user confidence, prevents unnecessary concern, and ensures that merchants can continue making business decisions using the available data.